OptoFoton — Photon Field in TauTimp Framework (Cylindrical Coordinates)

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1. Statement of the Problem

We study the TauTimp photon field equation in cylindrical coordinates with axial symmetry ($\partial_\theta = 0$):

$$\frac{2m}{\hbar^2}\left[E + \frac{e^2}{4\pi\varepsilon_0 r} - \frac{l(l+1)\hbar^2}{2mr^2}\right] H(r,z,t) + \frac{R}{r}\,\Delta H(r,z,t) = \frac{1}{c^2}\frac{\partial^2 H}{\partial t^2},$$

where the Laplacian in cylindrical coordinates with axial symmetry reads

$$\Delta H = \frac{\partial^2 H}{\partial r^2} + \frac{1}{r}\frac{\partial H}{\partial r} + \frac{\partial^2 H}{\partial z^2}.$$

Domain: $r > 0$, $z \in (-\infty, \infty)$, $t \in \mathbb{R}$. **Boundary condition:** $H(R,z,t) = 0$ (cosmological cutoff). **Regularity:** $H_3(r)$ finite as $r \to 0^+$. **Parameters:** $l \in \{0,1,2,3\}$ (free azimuthal-like quantum number), m is the photon effective mass scale in the TauTimp prefactor (kept symbolic).

2. Separation of Variables

We set

$$H(r,z,t) = H_3(r)\,H_4(z)\,u(t).$$

Substituting in (1) and dividing by $H_3 H_4 u$:

$$\frac{2m}{\hbar^2}\left[E + \frac{e^2}{4\pi\varepsilon_0 r} - \frac{l(l+1)\hbar^2}{2mr^2}\right] + \frac{R}{r}\left[\frac{H_3''}{H_3} + \frac{1}{r}\frac{H_3'}{H_3} + \frac{H_4''}{H_4}\right] = \frac{1}{c^2}\frac{\ddot{u}}{u}.$$

2.1 Temporal part

$$\frac{\ddot{u}}{u} = -\omega^2 \quad \Rightarrow \quad u(t) = e^{-i\omega t}.$$

2.2 Axial part (case (a): unbounded z, Fourier mode)

$$\frac{H_4''}{H_4} = -k_z^2 \quad \Rightarrow \quad H_4(z) = e^{ik_z z}.$$

2.3 Radial equation (variable-coefficient ODE)

After moving ω^2/c^2 and the k_z^2 term across, the radial equation becomes:

$$\boxed{\frac{R}{r}\left[H_3''(r) + \frac{1}{r}H_3'(r)\right] + \left\{\frac{2mE}{\hbar^2} + \frac{2m\,e^2}{4\pi\varepsilon_0\hbar^2 r} - \frac{l(l+1)}{r^2} - \frac{R\,k_z^2}{r} + \frac{\omega^2}{c^2}\right\} H_3(r) = 0}$$

Multiplying (6) by r/R:

$$H_3'' + \frac{1}{r}H_3' + \left\{\frac{r}{R}\left[\frac{2mE}{\hbar^2} + \frac{\omega^2}{c^2}\right] + \frac{2m\,e^2}{4\pi\varepsilon_0\hbar^2 R} - \frac{l(l+1)}{rR} - k_z^2\right\} H_3 = 0.$$

This is a **variable-coefficient linear ODE** with three regimes (linear-in-r, constant, and $1/r$ singular terms).

3. Reduction Strategies

Define the auxiliary constants:

$$\alpha \equiv \frac{1}{R}\left(\frac{2mE}{\hbar^2} + \frac{\omega^2}{c^2}\right), \quad \beta \equiv \frac{2me^2}{4\pi\varepsilon_0\hbar^2 R} - k_z^2, \quad \gamma \equiv \frac{l(l+1)}{R}.$$

The ODE compactly reads:

$$H_3'' + \frac{1}{r}H_3' + \left[\alpha\, r + \beta - \frac{\gamma}{r}\right] H_3 = 0.$$

3.1 Substitution to absorb the first-derivative term

Let $H_3(r) = r^{-1/2}\chi(r)$. Then:

$$H_3' = -1/2\, r^{-3/2}\chi + r^{-1/2}\chi',$$
$$H_3'' = 3/4\, r^{-5/2}\chi - r^{-3/2}\chi' + r^{-1/2}\chi''.$$

Substituting into (9):

$$\chi'' + \left[\alpha r + \beta - \frac{\gamma}{r} + \frac{1}{4r^2}\right]\chi = 0.$$

This is a **generalized Coulomb–linear–centrifugal** Schrödinger-like equation.

3.2 Limiting regimes

(i) Large-r regime ($r \to R$, αr dominates): equation reduces to

$$\chi'' + \alpha r\,\chi = 0,$$

which is the **Airy equation** under $\xi = -(\alpha)^{1/3} r$:

$$\frac{d^2\chi}{d\xi^2} - \xi\,\chi = 0 \quad \Rightarrow \quad \chi(r) = c_1\,\mathrm{Ai}\left[-\alpha^{1/3} r\right] + c_2\,\mathrm{Bi}\left[-\alpha^{1/3} r\right].$$

(ii) Constant-coefficient regime (β dominates, αr and γ/r negligible): equation (9) becomes the **Bessel equation of order zero**:

$$H_3'' + \frac{1}{r}H_3' + \beta H_3 = 0 \quad \Rightarrow \quad H_3(r) = A\,J_0(\sqrt{\beta}\,r) + B\,Y_0(\sqrt{\beta}\,r).$$

Regularity at $r = 0^+$ forces $B = 0$.

(iii) Small-r regime (γ/r dominates): equation (10) gives

$$\chi'' - \frac{\gamma}{r}\chi = 0 \quad \Rightarrow \quad \chi(r) \sim r^{1/2} I_1(2\sqrt{\gamma r}),$$

with I_1 the modified Bessel function (selecting the regular branch).

3.3 Quantization from the boundary condition $H_3(R) = 0$

Using the Bessel asymptote (13), the eigenvalue condition is:

$$J_0(\sqrt{\beta}\, R) = 0 \quad \Rightarrow \quad \sqrt{\beta_n}\, R = j_{0,n},$$

where $j_{0,n}$ is the n-th positive zero of J_0:

n	$j_{0,n}$
1	2.4048
2	5.5201
3	8.6537
4	11.7915

Thus the quantized dispersion relation:

$$\beta_n = \frac{j_{0,n}^2}{R^2} \quad \Rightarrow \quad \frac{2me^2}{4\pi\varepsilon_0\hbar^2 R} - k_z^2 = \frac{j_{0,n}^2}{R^2}.$$

For the dominant Airy branch, the condition $\mathrm{Ai}(-\alpha^{1/3}R) = 0$ yields the alternative quantization:

$$\alpha^{1/3}R = -a_n \quad (a_n \text{ are zeros of Ai}, a_1 = 2.3381, a_2 = 4.0879, \dots),$$

leading to $\frac{2mE_n}{\hbar^2} + \frac{\omega_n^2}{c^2} = \frac{a_n^3}{R^2}$.

4. Full Wave Function

Combining the separated solutions:

$$\boxed{H_{n,k_z,l}(r,z,t) = \mathcal{N}\, r^{-1/2}\, \mathrm{Ai}\left[-\alpha^{1/3} r\right] e^{ik_z z}\, e^{-i\omega_n t},}$$

with $\mathcal{N}$ a normalization constant and the constraint (18) tying ω_n, E_n, k_z, and l.

A useful approximate form valid in the bulk (r moderate, β-dominated): $H_{n,k_z}(r,z,t) \approx \mathcal{N}\, J_0\left(\frac{j_{0,n}}{R}\, r\right) e^{ik_z z}\, e^{-i\omega_n t}$.

5. Probability Density Wave Equation

We now study the d'Alembert equation for the probability density: $\frac{\partial^2 P}{\partial t^2} - c^2 \frac{\partial^2 P}{\partial r^2} = 0,$

with cylindrical symmetry, boundary condition $P(R, 0, 0) = 0$, and an **independent Gaussian ansatz** at $t = 0$ (not derived from $|\psi|^2$).

5.1 Gaussian ansatz at $t = 0$ (stationary snapshot)

$$\boxed{P(r,z,0) = A \exp\left[-\frac{(r-r_0)^2}{2\sigma_r^2}\right] \exp\left[-\frac{z^2}{2\sigma_z^2}\right],}$$

with A a normalization amplitude, $r_0 \in (0, R)$ the peak location, and σ_r, σ_z widths.

5.2 Boundary condition $P(R, 0, 0) = 0$

Strictly, a Gaussian never vanishes. We impose **effective vanishing** by requiring:

$$\frac{(R - r_0)^2}{2\sigma_r^2} \gg 1 \quad \text{e.g.,} \quad R - r_0 \geq 5\sigma_r.$$

A concrete choice consistent with the TauTimp scale:

$$r_0 = R/2, \quad \sigma_r = R/10, \quad \sigma_z = R/10, \quad A = 1.$$

This gives $P(R, 0, 0) = e^{-12.5} \approx 3.7 \times 10^{-6} \approx 0$ at the cutoff.

5.3 D'Alembert solution for $t > 0$

The general solution of (21) is: $P(r,z,t) = f(r - ct, z) + g(r + ct, z)$.

For the symmetric initial profile (22) with $\partial_t P|_{t=0} = 0$ (stationary initial condition):

$$\boxed{P(r,z,t) = \frac{A}{2}\left\{\exp\left[-\frac{(r - ct - r_0)^2}{2\sigma_r^2}\right] + \exp\left[-\frac{(r + ct - r_0)^2}{2\sigma_r^2}\right]\right\} \exp\left[-\frac{z^2}{2\sigma_z^2}\right].}$$

This represents two counter-propagating Gaussian wavepackets in r, with the z-envelope unchanged.

6. Maxwell Correspondence (Option A — Formal)

Equation (21) is the radial reduction of the vacuum d'Alembert equation:

$$\Box\phi \equiv \left(\frac{1}{c^2}\partial_t^2 - \nabla^2\right)\phi = 0.$$

Each Cartesian component of the electric field $\boldsymbol{E}$, magnetic field $\boldsymbol{B}$, vector potential $\boldsymbol{A}$ (in Lorenz gauge), and scalar potential ϕ satisfies (27) in vacuum:

$$\Box E_i = 0, \quad \Box B_i = 0, \quad \Box A_\mu = 0.$$

Dispersion relation: plane-wave ansatz $\phi \propto e^{i(kr - \omega t)}$ gives:

$$\omega^2 = c^2 k^2 \quad \Rightarrow \quad \omega = ck.$$

Thus the TauTimp probability density $P = |\psi|^2$ propagates as a **Maxwell-type scalar mode** with the universal light-cone dispersion. The 1D radial form $(\partial_t^2 - c^2\,\partial_r^2)P = 0$ is the reduction obtained when transverse (z, θ) dependence is frozen, i.e., when P depends on r alone up to a factorizable envelope.

7. Numerical Parameter Summary

Quantity	Symbol	Value
Cosmological radius	R	3.9228×10^{22} m
Speed of light	c	2.998×10^{8} m/s
	$\hbar$	1.0546×10^{-34} J·s
Vacuum permittivity	ε_0	8.854×10^{-12} F/m
Elementary charge	e	1.602×10^{-19} C
Free parameters	l	$\{0,1,2,3\}$
First J_0 zero	$j_{0,1}$	2.4048
First Airy zero	a_1	2.3381
Gaussian peak	r_0	$R/2 = 1.9614 \times 10^{22}$ m
Radial width	σ_r	$R/10 = 3.9228 \times 10^{21}$ m
Axial width	σ_z	$R/10 = 3.9228 \times 10^{21}$ m

Appendix B — Explicit Maxwell Field Mapping

B.1 TauTimp ψ as a component of the vector potential

We identify the TauTimp wavefunction with a Cartesian component of $\boldsymbol{A}$ in Lorenz gauge:

$$\psi(r, z, t) \equiv A_z(r, z, t),$$

so A_z satisfies (27) and admits the cylindrical mode (19)–(20).

B.2 Field strengths

In cylindrical coordinates with $\boldsymbol{A} = A_z(r, z, t)\,\hat{\boldsymbol{z}}$:

$$\boldsymbol{B} = \nabla \times \boldsymbol{A} = -\frac{\partial A_z}{\partial r}\,\widehat{\boldsymbol{\theta}}, \quad \boldsymbol{E} = -\frac{\partial \boldsymbol{A}}{\partial t} = -\frac{\partial A_z}{\partial t}\,\hat{\boldsymbol{z}}.$$

For the mode (20):

$$B_\theta = -\frac{\partial}{\partial r}\left[\mathcal{N} J_0(k_n r) e^{i(k_z z - \omega_n t)}\right] = \mathcal{N} k_n J_1(k_n r)\, e^{i(k_z z - \omega_n t)},$$

$$E_z = i\omega_n \mathcal{N} J_0(k_n r)\, e^{i(k_z z - \omega_n t)},$$

with $k_n = j_{0,n}/R$.

B.3 Energy density

The electromagnetic energy density:

$$u_{\mathrm{EM}} = \frac{\varepsilon_0}{2}|\boldsymbol{E}|^2 + \frac{1}{2\mu_0}|\boldsymbol{B}|^2.$$

For the TM-like mode above:

$$u_{\mathrm{EM}}(r, z, t) = \frac{\varepsilon_0}{2}\omega_n^2 |\mathcal{N}|^2 J_0^2(k_n r) + \frac{1}{2\mu_0} k_n^2 |\mathcal{N}|^2 J_1^2(k_n r).$$

B.4 Connection to $P = |\psi|^2$

By identification (B.1): $P(r, z, t) = |\psi|^2 = |A_z|^2 = |\mathcal{N}|^2 J_0^2(k_n r)$.

In the radiation-zone limit ($J_0^2 \approx J_1^2$ on average over a cycle), and using $\omega_n = c k_n$ in vacuum:

$$\frac{\varepsilon_0}{2}\omega_n^2 = \frac{\varepsilon_0 c^2 k_n^2}{2} = \frac{k_n^2}{2\mu_0},$$

so the electric and magnetic contributions are equal, and:

$$\boxed{u_{\mathrm{EM}} \propto |\psi|^2 = P \quad \text{(up to } \omega_n^2 \text{ and } k_n^2 \text{ prefactors).}}$$

This establishes the explicit identification of the TauTimp probability density with the EM energy density of the photon field mode, completing the Maxwell correspondence.

B.5 Poynting flux

$$\boldsymbol{S} = \frac{1}{\mu_0}\boldsymbol{E} \times \boldsymbol{B} = \frac{1}{\mu_0}(-\hat{\boldsymbol{z}}\,\partial_t A_z) \times (-\widehat{\boldsymbol{\theta}}\,\partial_r A_z) = \frac{1}{\mu_0}(\partial_t A_z)(\partial_r A_z)\,\hat{\boldsymbol{r}}.$$

The radial flux confirms outward photon propagation, consistent with the d'Alembert outgoing-wave branch in (26).

Summary of Deliverables

- **Section 1–4:** Derivation of the variable-coefficient photon wave equation, separation of variables, Bessel/Airy reduction, eigenvalue quantization, and full wavefunction.
- **Section 5:** Independent Gaussian probability density at $t = 0$, boundary condition handling, and d'Alembert evolution as two counter-propagating Gaussians.
- **Section 6:** Maxwell correspondence (formal, Option A).
- **Appendix B:** Explicit field-mapping correspondence with EM energy density.
- **OptoFoton.mpl:** Maple transcript with symbolic derivation, eigenvalue computation, and 5 plot panels.